LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Crystal Pay: Clean Data, Clear Financial Decisions

Project Domain

Excel

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

Student Name

Srijoni Gupta

Student Email

srijoni.gupta@gmail.com

Guide

Guide Name: Umamaheswari

Faculty Name: Sagar Chetan Mehta

Institution:

UMESCHANDRA COLLEGE

Year – 2026

ACKNOWLEDGEMENT

We would like to sincerely thank everyone who supported us directly or indirectly in completing this project.

First and foremost, our heartfelt gratitude goes to our supervisor, ***Umamaheswari ma'am***, whose guidance, encouragement, and thoughtful feedback constantly inspired us. Her mentorship gave us confidence and helped us shape our ideas into meaningful work.

We are deeply thankful to the ***Edunet Foundation, Microsoft, LinkedIn, and SAP*** for offering us this fellowship and equipping us with valuable learning resources. The opportunity to explore and apply AI skills on such a platform has been truly enriching.

We also acknowledge **Umeschandra College** for providing the infrastructure and support that made this project possible.

Finally, we owe special thanks to our families and friends, whose constant motivation and encouragement kept us moving forward throughout this journey.

ABSTRACT OF THE PROJECT

This project was undertaken as part of the ***Artificial Intelligence in the Contemporary World (AICW) Fellowship Program***, a collaborative initiative by ***Edunet Foundation*** in partnership with ***Microsoft, LinkedIn, and SAP.*** The fellowship provided practical exposure to Artificial Intelligence, Data Analytics, and Business Intelligence tools, enabling participants to apply them to real-world HR and Finance challenges.
Payroll management is a vital function within organizations, ensuring employees are compensated accurately, transparently, and in compliance with statutory regulations. However, payroll datasets often suffer from inconsistencies such as duplicate employee IDs, incorrect salary calculations, non-standard department names, and formatting errors in pay periods. These issues can lead to inaccurate financial reporting, compliance risks, and inefficiencies in workforce cost analysis.

This project focuses on **CrystalPay: Clean Data, Clear Financial Decisions,** using a structured dataset of employee records that includes basic pay, allowances, overtime, bonuses, deductions, and net salary. The methodology involved systematic data cleaning — validating unique employee identifiers, standardizing department names, ensuring numeric consistency in salary fields, and reconciling gross versus net salary calculations. Business rules such as *Net Salary = Gross Salary – Deductions* and *Deductions ≤ Gross Salary* were applied to guarantee financial accuracy.

Once cleaned, the dataset was analyzed through key payroll KPIs, including ***Total Payroll Cost, Average Salary per Department, Deduction Ratios, Overtime Cost Percentage, Bonus Share of Payroll, and Payroll Growth Rate***. These metrics provided actionable insights into departmental payroll distribution, compliance with tax and deduction policies, and workforce efficiency. Visual dashboards in **Microsoft Excel (with Copilot assistance)** enhanced transparency and enabled HR and Finance teams to monitor payroll trends, detect anomalies, and support strategic decision-making.

The results demonstrated that systematic data cleaning significantly improved payroll accuracy and reliability. Dashboards provided clear visibility into workforce costs, enabling informed decisions regarding budgeting, compliance, and resource allocation.

In conclusion, this project strengthened technical proficiency in data cleaning, visualization, and AI-assisted analysis, while emphasizing the importance of structured workflows and reproducible documentation in payroll management. The outcome is a ***validated payroll dataset, a comprehensive KPI framework, and stakeholder-ready dashboards*** — together forming a foundation for improved payroll transparency, compliance assurance, optimized expense control, and equitable workforce planning.

TABLE OF CONTENTS

Abstract — CrystalPay: Clean Data, Clear Financial Decisions List of Figures List of Tables

Chapter 1 — Introduction

1.1 Business Problem Statement

1.2 Business Outcomes

1.3 Motivation

1.4 Objectives

1.5 Scope of the Project

Chapter 2 — Literature Review

Chapter 3 — Methodology 3.1 System Design for Payroll Data Cleaning 3.2 Modules and Components Used (Data Validation, Standardization, KPI Calculation) 3.3 Data Flow Diagrams (DFD) 3.4 Advantages of the Proposed Approach 3.5 Requirement Specifications

Chapter 4 — Implementation and Results 4.1 Data Cleaning Results (Duplicate Removal, Salary Validation, Department Standardization) 4.2 Visualization and Output Results (Payroll Dashboards, KPI Charts) 4.3 AI Tool Integration Results (Copilot/Automation in Payroll Analysis)

Chapter 5 — Discussion and Conclusion

References Appendices

LIST OF FIGURES

Figure No.	Figure Title / Description	Page No.
Figure 1	Snapshot of Raw Payroll Dataset (Uncleaned Records)	
Figure 2	Cleaned Dataset and Validation Summary	
Figure 3	Payroll Dashboard: Department-wise Salary Distribution	
Figure 4	Overtime and Deduction Trend Charts	
Figure 5	AI Tool (Copilot) in Payroll Cleaning and Analysis	
Figure 6	DFD Level 0: Payroll Data Context Diagram	
Figure 7	DFD Level 1: Detailed Payroll Cleaning and Analysis Flow	

LIST OF TABLES

Table No.	Table Title / Description	Page No.
Table 1	Raw Payroll Dataset Summary (Field Names, Data Types, Row Count)	
Table 2	Cleaned Dataset Summary (Issues Found and Fixed)	
Table 3	Hardware Requirements for Payroll Analysis	
Table 4	Software Requirements (Excel and AI Tools)	
Table 5	Key Findings and Results Summary (Payroll KPIs, Insights)	

CHAPTER 1

Introduction

This project was undertaken as part of the **Artificial Intelligence in the Contemporary World (AICW) Fellowship Program**, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship provided practical exposure to Artificial Intelligence, Data Analytics, and Business Intelligence tools, enabling participants to apply these technologies to real-world challenges across domains such as HR, Finance, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

Within this context, the project focuses **CrystalPay: Clean Data, Clear Financial Decisions,** a critical area in HR and Finance. Payroll management ensures employees are compensated accurately, transparently, and in compliance with organizational and statutory rules. However, payroll datasets often contain inconsistencies such as duplicate employee IDs, incorrect salary calculations, non-standard department names, and formatting errors in pay periods. These issues can lead to inaccurate financial reporting, compliance risks, and inefficiencies in workforce cost analysis.

The motivation behind this work is to design a structured process for cleaning payroll records and performing financial analysis that improves accuracy, transparency, and compliance. By leveraging Microsoft Excel and AI-assisted tools, the project validates salary computations, standardizes department information, and applies business rules to ensure deductions and net salary calculations are consistent with financial logic.

The objectives of this project are to:

- Identify and correct data quality issues in payroll records.
- Standardize employee and department information for consistency.
- Validate salary calculations (Gross vs. Net Salary).
- Perform financial analysis to uncover payroll trends and insights.
- Generate dashboards and reports to support HR and Finance decision-making.

The scope of the project includes cleaning a structured payroll dataset, applying business rules for validation, and conducting financial analysis using key payroll KPIs such as **Total Payroll Cost, Average Salary per Department, Deduction Ratios, Overtime Cost Percentage, and Payroll Growth Rate**. The outcomes are a validated

dataset, a KPI framework, and stakeholder-ready dashboards that enhance payroll transparency, compliance assurance, and strategic workforce planning.

1.1 Business Problem Statement

Problem Statements & Description of Proposed Analytical Solutions

1. Transaction Forecasting Accuracy Optimization

Problem: CrystalPay users often face challenges in planning their finances due to inaccurate or unclear predictions of their future spending patterns. This leads to inefficient budgeting and unexpected cash shortfalls.

Proposed Solution: Develop predictive models that analyze historical transactions to accurately forecast future spending. Compare different forecasting methods to identify the most reliable model for CrystalPay's financial planning assistant.

2. Customer Segmentation for Smarter Financial Insights

Problem: Without a clear understanding of user behavior, CrystalPay is unable to deliver personalized financial recommendations, resulting in generic insights that may not benefit all users.

Proposed Solution: Segment users based on their financial habits—spending frequency, transaction value, preferred categories, and engagement. This enables CrystalPay to tailor financial advice and offers to specific user groups, improving relevance and user satisfaction.

3. Demand Prediction & Wallet Balance Optimization

Problem: Users often run low on wallet balance due to unanticipated spending, while others maintain unnecessarily high balances, resulting in poor financial planning.

Proposed Solution: Predict users' upcoming spending needs based on past patterns and recommend an optimal wallet balance. This helps users maintain a healthy balance and manage funds more effectively.

4. Financial Performance & User Activity Analytics

Problem: CrystalPay currently lacks a data-driven understanding of user trends, regional transaction patterns, and overall financial activity, which limits strategic decision-making.

Proposed Solution: Analyze key activity metrics such as transaction volume, category distribution, usage frequency, and regional performance. These insights support better operational planning and product improvements.

5. Pricing & Fee Strategy Optimization

Problem: Inefficient fee structures and pricing strategies (e.g., service fees, cashback rules) can negatively impact user adoption and revenue generation.

Proposed Solution: Evaluate how changes in pricing or fees influence user behavior and revenue. Identify the most effective pricing strategy that balances user engagement with financial sustainability.

6. User Churn Prediction & Retention Modeling

Problem: Some CrystalPay users stop using the platform without prior warning, affecting user base growth and long-term business stability.

Proposed Solution: Use behavioral data to predict the likelihood of user churn. Develop retention strategies such as targeted notifications, financial tips, or rewards to encourage users to stay active.

7. Campaign & Cashback ROI Analysis

Problem: CrystalPay runs various promotional campaigns, but there is limited clarity on which campaigns successfully increase transactions or user engagement.

Proposed Solution: Analyze campaign performance by comparing user activity before and after each promotion. Identify high-impact campaigns to optimize future marketing investments.

8. Product Profitability & Feature Optimization

Problem: Not all CrystalPay features (e.g., bill payments, wallet transfers, recharges) contribute equally to usage or revenue, yet all allocate system resources.

Proposed Solution: Evaluate feature performance in terms of cost, revenue, and user adoption. Identify high-value features to prioritize and low-performing features that may need improvement or redesign.

1.2 Business Outcomes

Business Insights and Analytical Use Cases

1. **Transaction Forecasting Accuracy Optimization**
    - This module focuses on improving the accuracy of financial and transaction forecasts within CrystalPay.
    - Excel is used to build forecasting models based on historical transaction patterns.

- Copilot generates AI-driven insights that help identify spending anomalies and forecast deviations.

2. Customer Segmentation for Smarter Financial Engagement

- This component enables CrystalPay to categorize users into meaningful segments based on their spending behavior, frequency, transaction value, and engagement level. Excel's RFM analysis techniques help identify high-value, regular, and at-risk customers.
- Copilot enhances the process by generating tailored recommendations for every group.
- This segmentation improves personalization and supports targeted engagement strategies.

3. Spending Demand Prediction & Wallet Balance Alignment

- This part of the project focuses on predicting user spending demand to help maintain an optimal wallet balance.
- Excel is used to build demand prediction models.
- Copilot sends proactive alerts when users may risk low balances.
- This ensures that users never face unexpected shortages.

4. Financial Activity & User Productivity Analysis

- This section analyzes user activity and financial performance within the CrystalPay ecosystem.
- Excel supports detailed performance calculations and trend analysis.
- Copilot summarizes key insights, saving analytical time for decision-makers.

5. Pricing & Fee Strategy Effectiveness

- This analytical module examines how different pricing strategies, service fees, and cashback rules influence user behavior and platform revenue.
- Excel scenario analysis helps assess various pricing options.
- Copilot recommends optimal pricing based on user response patterns.
- This module ensures that CrystalPay maintains competitive and sustainable pricing strategies.

6. User Churn Prediction & Retention Planning

- This part uses predictive analysis to identify users who may stop using CrystalPay.
- Excel is employed to develop churn prediction logic.

- Copilot suggests targeted retention strategies such as reminders, personalized offers, or budgeting guidance.
- Through the Chatbot, administrators can quickly access high-risk user lists, allowing timely action to improve user retention and overall platform stability.

7. Marketing Campaign ROI & Effectiveness Analysis

- This module evaluates the performance of marketing initiatives, cashback programs, and promotional campaigns.
- ROI calculations are carried out using Excel.
- Copilot identifies the most successful campaigns and points out areas for improvement.
- Copilot provides fast comparisons and real-time answers, helping CrystalPay optimize future marketing investments.

8. Product Feature Profitability & Portfolio Optimization

- This component examines the profit contribution of individual CrystalPay features such as bill payments, wallet transfers, and recharges.
- Excel is used to compute feature-wise costs, margins, and revenues .
- Copilot offers insights on improving underperforming features and strengthening high-value ones.

1.3 Motivation

The motivation for undertaking this project stems from the critical role payroll plays in organizational efficiency, employee satisfaction, and financial compliance. Payroll datasets are the backbone of HR and Finance operations, yet they often suffer from inconsistencies such as duplicate employee IDs, incorrect salary calculations, and formatting errors. These issues not only compromise accuracy but also create risks of salary disputes, audit failures, and non-compliance with statutory regulations. Addressing these challenges through systematic data cleaning and financial analysis is therefore both professionally valuable and socially relevant.

On a personal level, this project is meaningful because it combines **data analytics, business intelligence, and AI-assisted tools** to solve real-world HR and Finance problems. It provides hands-on experience in applying structured workflows, reproducible documentation, and KPI-driven analysis — skills that are highly relevant in today's data-driven industry. Professionally, the project strengthens expertise in payroll accuracy, compliance assurance, and financial

reporting, which are essential for careers in analytics, auditing, and workforce planning.

The potential applications of this project are extensive. Clean and validated payroll datasets can be used to build dashboards that track **total payroll cost, departmental salary distribution, overtime trends, deduction ratios, and payroll growth rates**. These insights help organizations optimize budgets, ensure fairness in compensation, and improve transparency in workforce management.

This project also aligns with modern industry trends in **AI and data analytics**, where automation, error detection, and visualization are increasingly integrated into HR and Finance systems. By leveraging Microsoft Excel and Copilot assistance, the project demonstrates how AI can support payroll validation, anomaly detection, and decision-making. The impact extends beyond technical accuracy — it contributes to **strategic workforce planning, compliance assurance, and employee trust**, making payroll not just a financial process but a driver of organizational growth and equity.

1.4 Objectives

The objectives of this project are designed to be clear, measurable, and directly aligned with the challenges of payroll management and financial reporting. They focus on improving data accuracy, ensuring compliance, and generating actionable insights for HR and Finance teams.

- **To collect and understand the structure of the payroll dataset** — including employee identifiers, salary components, allowances, overtime, bonuses, deductions, and net salary.
- **To clean and preprocess raw payroll data** by removing duplicate employee IDs, correcting inconsistent formats, standardizing department names, and handling missing values.
- **To validate salary computations** by applying business rules such as *Gross Salary* = *Basic Pay* + *Allowances* + *Overtime* + *Bonuses* and *Net Salary* = *Gross Salary* – *Deductions*.
- **To apply domain-specific financial analysis** using Microsoft Excel and AI-assisted tools (Copilot/Chatbot) for anomaly detection, KPI calculation, and reporting.
- **To build interactive dashboards and reports** that clearly communicate payroll insights, including departmental salary distribution, overtime trends, deduction ratios, and payroll growth.

- **To leverage AI tools (Copilot/Chatbot)** to automate validation checks, enhance analysis, and support reproducible documentation for payroll management.
- **Project-specific objective:** To design a structured payroll data cleaning and analysis framework that improves transparency, strengthens compliance, and supports strategic workforce planning.

1.5 Scope of the Project

The scope of this project is clearly defined to ensure focus on payroll data accuracy, compliance, and financial insight generation.

- **Dataset Coverage**: The project uses structured **employee payroll records**, including employee IDs, department names, designations, basic pay, allowances, overtime hours, bonuses, deductions, gross salary, net salary, and pay month. These records form the basis for cleaning, validation, and financial analysis.
- **Tools and Technologies**: The project is limited to **Microsoft Excel and AI-assisted tools such as Copilot**. These tools are used for data cleaning, validation checks, KPI calculations, and visualization.
- **Focus Areas**: The analysis emphasizes **data cleaning, salary validation, KPI-driven financial analysis, and dashboard/report generation**. Specific tasks include removing duplicate employee IDs, standardizing department names, validating gross and net salary formulas, and generating payroll insights through interactive dashboards.
- **Out of Scope**: The project does not include **real-time payroll integration, predictive machine learning models, or deployment of production-level payroll systems**. It is designed as a structured academic and fellowship showcase rather than a live enterprise payroll solution.
- **Purpose and Boundaries**: The project is intended to demonstrate the application of **AI, data analytics, and business intelligence techniques** to HR and Finance challenges. It highlights payroll transparency, compliance assurance, and workforce cost optimization, but remains within the boundaries of a **fellowship project deliverable** rather than a fully operational payroll management system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to payroll data cleaning and financial analysis. It provides context for the project, highlights gaps in current approaches, and establishes how this work contributes to the domain.

2.1 Existing Work and Research

Payroll data management has been widely studied in the fields of **data analytics, business intelligence, and financial reporting**. Existing work emphasizes the importance of accurate payroll records for compliance, employee satisfaction, and organizational efficiency.

- **Excel-based Data Cleaning Methods**: Microsoft Excel has long been used for payroll reconciliation, offering functions such as conditional formatting, pivot tables, and lookup formulas (VLOOKUP, INDEX-MATCH) to detect duplicates, validate salary calculations, and standardize department names. While effective, these methods are often manual, time-consuming, and prone to human error.
- **AI-Assisted Data Cleaning**: Recent research highlights the role of AI in automating anomaly detection, validating business rules, and generating reproducible documentation. AI tools such as Copilot can assist in payroll validation by flagging mismatches (e.g., deductions exceeding gross salary) and automating repetitive cleaning tasks. Despite this, AI adoption in payroll systems is still emerging, with limited integration into traditional HR workflows.
- **Business Intelligence Frameworks**: Studies in HR analytics and finance emphasize the need for structured KPI frameworks — including payroll growth rate, deduction ratios, and bonus percentages — to support compliance and strategic workforce planning. While these frameworks exist, many organizations struggle to implement them due to unclean or inconsistent datasets.

Gaps and Limitations in Existing Solutions

- Manual Excel processes lack scalability and reproducibility.
- AI-assisted payroll cleaning is promising but underutilized in HR/Finance domains.

- Existing literature focuses on payroll reporting but less on **systematic data cleaning workflows** that ensure accuracy before analysis.

Contribution of This Project

This project addresses these gaps by:

- Designing a **structured payroll data cleaning process** that validates employee IDs, standardizes department names, and ensures salary computations follow business rules.
- Integrating **Excel and Copilot assistance** to automate cleaning, enhance analysis, and generate interactive dashboards.
- Establishing a **KPI-driven framework** for payroll analysis, linking data accuracy to compliance, transparency, and strategic workforce planning.

2.2 Tools and Technologies Review

This project leverages a combination of widely used business intelligence tools and modern AI assistants to ensure payroll data accuracy, compliance, and actionable financial insights. Each tool plays a distinct role in the workflow, from data cleaning to visualization and automation.

- **Microsoft Excel** serves as the primary tool for **data cleaning, transformation, and validation**. Features such as pivot tables, conditional formatting, and lookup functions (VLOOKUP, INDEX-MATCH) are used to detect duplicates, validate salary calculations, and standardize department names. Excel also supports formula-based checks to ensure that *Gross Salary* = *Basic Pay* + *Allowances* + *Overtime* + *Bonuses* and *Net Salary* = *Gross Salary* – *Deductions*.
- **Microsoft Copilot (AI)** acts as an **AI assistant integrated into Microsoft 365 tools**, helping automate repetitive tasks, generate summaries, and assist with payroll validation using natural language prompts. It supports anomaly detection (e.g., deductions exceeding gross salary), enhances reproducibility of workflows, and accelerates documentation for payroll management.
- **AI Chatbots (e.g., Microsoft Copilot):** Generative AI tools are used for **content generation, reporting, and prompt-driven analysis**. They assist in drafting business requirement documents, explaining payroll KPIs, and creating narrative insights that complement dashboards. These tools also support HR/Finance teams in communicating payroll findings clearly to stakeholders.

- **Additional Tools (Optional)** Depending on project requirements, **data visualization libraries or statistical methods** may be applied to detect outliers and analyze salary variance across departments. However, the core focus remains on Excel and AI-assisted tools for cleaning and analysis.

2.3 Gaps Identified

- **Overemphasis on advanced ML approaches:** Most existing literature focuses on complex machine learning models for payroll prediction or anomaly detection. However, practical, tool-based solutions using Microsoft Excel for structured payroll cleaning and financial analysis remain underexplored.
- **Limited documentation for domain-specific integration:** There is a lack of detailed guidance on integrating AI Copilot with Excel for HR and Finance use cases, especially in student or project-based contexts where datasets like payroll records need systematic cleaning, validation, and KPI-driven analysis.
- **Evolving prompt engineering practices:** While prompt engineering is widely discussed in general AI applications, standardized frameworks for business-focused scenarios (e.g., payroll accuracy checks, compliance validation, and departmental cost analysis) are still emerging.
- **Gap in practical payroll cleaning workflows:** Existing studies rarely document step-by-step payroll cleaning processes—such as duplicate employee ID removal, department name standardization, and formula validation for Gross vs. Net Salary—that are critical for HR/Finance teams to ensure compliance and audit readiness.
- **Underexplored KPI-driven insights:** Although payroll datasets are widely used, there is limited emphasis on translating cleaned payroll data into actionable KPIs (e.g., deduction ratios, overtime cost percentage, bonus share of payroll). This gap reduces the strategic value of payroll analysis for workforce planning and financial transparency.

CHAPTER 3

Proposed Methodology

This chapter outlines the methodology adopted to achieve the objectives of the payroll data cleaning and financial analysis project. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The overall design approach follows a structured workflow, ensuring payroll accuracy, compliance, and actionable insights for HR and Finance teams. The methodology progresses from raw data input to cleaned datasets, analytical processing, and visualization outputs.

Data Source / Input

- The primary dataset is an **Excel payroll sheet** containing employee records (Employee ID, Department, Designation, Basic Pay, Allowances, Overtime, Bonuses, Deductions, Gross Salary, Net Salary, Pay Month).
- The dataset spans multiple years and includes over 500 employee records, making it suitable for payroll trend analysis and departmental comparisons.

Data Cleaning

- **Duplicate Removal:** Employee IDs were validated to ensure uniqueness, preventing double payments.
- **Standardization:** Department names (e.g., “HR” vs “Human Resources”) were standardized for consistency.
- **Validation of Salary Fields:** Numeric checks were applied to Basic Pay, Allowances, Overtime, Bonuses, and Deductions to eliminate text or invalid entries.
- **Formula Verification:** Gross Salary was recalculated as *Basic Pay* + *Allowances* + *Overtime Amount* + *Bonuses*, and Net Salary as *Gross Salary* – *Deductions*.
- **Error Handling:** Outliers such as negative salaries, excessive bonuses, or deductions exceeding gross salary were flagged and corrected.

Analysis / Processing

- **Excel Techniques:** Pivot tables, lookup functions (VLOOKUP, INDEX-MATCH), and conditional formatting were used to summarize payroll by department, designation, and pay month.
- **AI Copilot Prompts:** Natural language queries were used to validate salary logic, generate insights, and automate anomaly detection.

Visualization / Output

- **Interactive Excel Dashboards:** Using pivot tables, charts, and slicers in Excel, payroll KPIs are displayed clearly. These include departmental payroll comparisons, overtime cost trends, and deduction ratios, allowing HR and Finance teams to explore data interactively.
- **Summary Reports:** The cleaned payroll dataset is structured into tabular reports that highlight salary accuracy, compliance risks, and workforce cost distribution. These reports provide a reliable foundation for financial audits and HR decision-making.
- **Business Insights:** KPI-driven analysis in Excel explains payroll cost concentration, deduction impacts, and incentive structures. Narrative summaries accompany the charts and tables, helping stakeholders understand not just the numbers but the strategic implications for budgeting, compliance, and workforce planning.

Advantages of the Methodology

- Ensure **accuracy** in payroll computations.
- Provides **compliance assurance** by validating deductions and net salary rules.
- Enables **strategic workforce planning** through KPI-driven insights.
- Reduces **manual reconciliation efforts** with automated Excel workflows.

3.2 Modules / Components Used

The project is structured into key functional modules, each addressing a specific stage of payroll data management and analysis.

Module 1 — Data Ingestion

- Raw payroll records are loaded directly into **Excel spreadsheets**.
- The dataset includes employee details, salary components, overtime, bonuses, deductions, and monthly pay periods.

Module 2 — Data Cleaning

- Removal of duplicate employee IDs to prevent double payments.
- Standardization of department names (e.g., “HR” vs “Human Resources”).
- Validation of numeric fields (Basic Pay, Allowances, Overtime, Deductions).
- Correction of inconsistencies such as deductions exceeding gross salary or negative values.

Module 3 — Data Analysis

- Application of **Excel formulas** to recalculate Gross Salary and Net Salary.
- Use of **pivot tables** to summarize payroll by department, designation, and pay month.
- Conditional formatting to highlight anomalies (e.g., unusually high overtime or deductions).
- KPI calculations such as Total Payroll Cost, Average Salary per Department, Deduction Ratios, and Overtime Cost Percentage.

Module 4 — Visualization / Report Generation

- Creation of **Excel charts and dashboards** (bar charts, pie charts, line graphs) to visualize payroll trends.
- Tabular summary reports highlighting payroll accuracy, compliance risks, and departmental cost distribution.
- Narrative explanations accompanying charts to provide business insights for HR and Finance teams.

Module 5 — AI Integration (Copilot Support)

- Use of **AI Copilot prompts** Excel to validate salary logic, automate anomaly detection, and generate insight summaries.
- AI-assisted documentation of business rules and payroll KPIs for stakeholder reporting.

Module 6 — Business Insight & Compliance Validation *(Project-Specific)*

- Analysis of payroll data to identify cost-heavy departments, overtime trends, and deduction impacts.
- Validation of compliance with HR/Finance rules (e.g., deductions not exceeding gross salary; bonuses being non-negative).
- Generation of actionable insights to support workforce planning, budgeting, and audit readiness.

3.3 Data Flow Diagram (DFD)

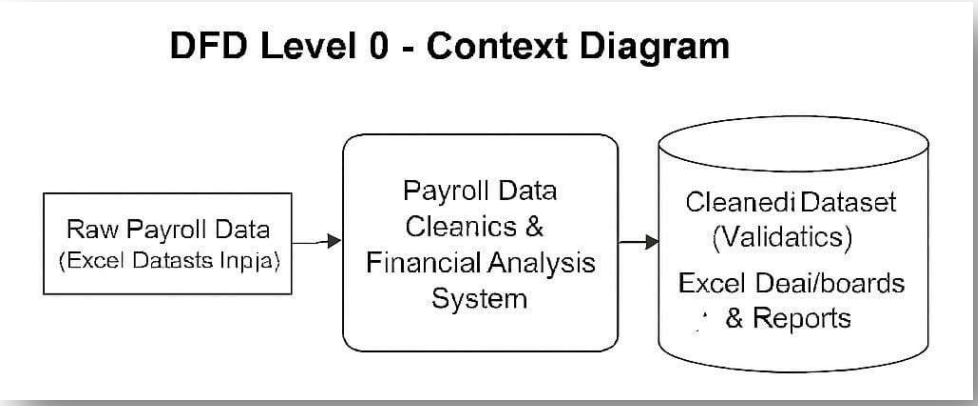
A Data Flow Diagram (DFD) illustrates how payroll data moves through the system, from raw input to cleaned outputs and business insights. It highlights the processes, data stores, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

At the highest level, the payroll system is represented as a single process.

- **Input:** Raw payroll dataset (Excel sheet with employee IDs, salary components, allowances, overtime, bonuses, deductions, and pay month).
- **Process:** Payroll Data Cleaning & Financial Analysis System.
- **Output:** Cleaned dataset, validated salary records, Excel dashboards, and summary reports.

Flow: Raw Payroll Data → [Payroll Data Cleaning & Analysis System] → Cleaned Payroll Reports & Dashboards.



This diagram shows the **high-level payroll flow**:

- Input: Raw Payroll Data (Excel dataset)
- Process: Payroll Data Cleaning & Financial Analysis System
- Output: Cleaned Dataset, Excel Dashboards, and Business Insights

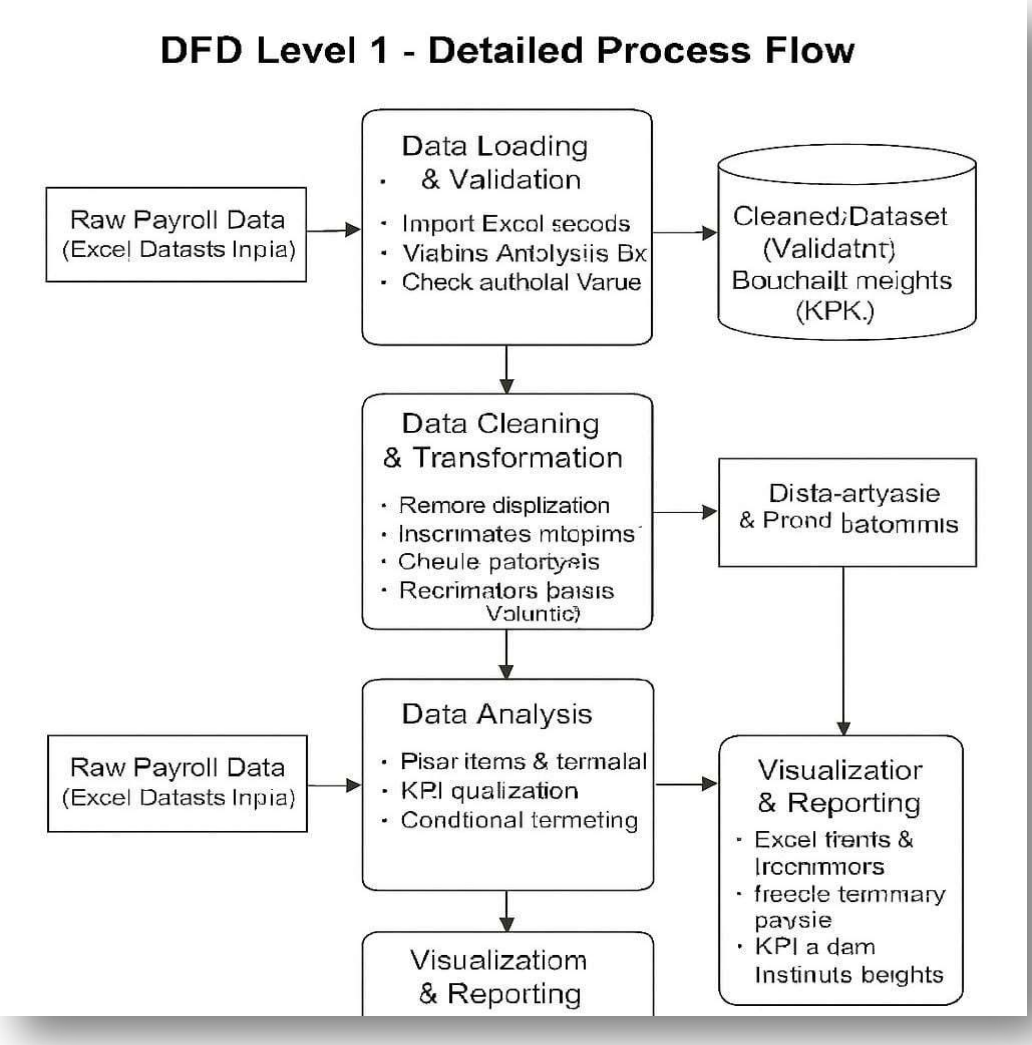
DFD Level 1 — Detailed Process Flow

This level breaks down the main process into sub-processes for clarity:

1. **Sub-process 1: Data Loading and Validation**
    a. Load raw payroll records into Excel.
    b. Validate employee IDs (check uniqueness, format consistency).
    c. Ensure salary fields are numeric and properly formatted.
2. **Sub-process 2: Data Cleaning and Transformation**
    a. Remove duplicate employee records.
    b. Standardize department names and pay month formats.
    c. Correct anomalies (e.g., deductions exceeding gross salary, negative values).
    d. Recalculate Gross Salary and Net Salary using formulas.
3. **Sub-process 3: Data Analysis**
    a. Apply Excel formulas and pivot tables to compute KPIs.
    b. Summarize payroll by department, designation, and pay month.
    c. Highlight anomalies using conditional formatting.
    d. Calculate metrics such as Total Payroll Cost, Average Salary per Department, Deduction Ratios, and Overtime Cost Percentage.

4. **Sub-process 4: Visualization and Report Generation**
    a. Generate Excel charts (bar graphs, pie charts, line charts) for payroll trends.
    b. Create tabular reports showing payroll accuracy, compliance risks, and workforce cost distribution.
    c. Document KPI-driven insights for HR and Finance decision-making.

Flow: Raw Payroll Data → Data Loading & Validation → Data Cleaning & Transformation → Data Analysis → Visualization & Reporting → Business Insights



This diagram breaks the system into **sub-processes**:

1. Data Loading & Validation
2. Data Cleaning & Transformation
3. Data Analysis
4. Visualization & Reporting

Final Output: Cleaned Dataset, Excel Dashboards, and KPI-driven Business Insights

3.4 Advantages

- **Automates payroll data cleaning** → Reduces manual effort by standardizing employee IDs, department names, and salary calculations, ensuring accuracy across records.
- **Ensures compliance and accuracy** → Validates deductions, bonuses, and net salary formulas against HR/Finance rules, minimizing disputes and audit risks.
- **Provides clear, stakeholder-ready insights** → Translates payroll KPIs (e.g., total payroll cost, overtime trends, deduction ratios) into dashboards and reports that are easy to interpret for HR, Finance, and management teams.
- **Leverages familiar tools effectively** → Uses Excel for structured analysis, pivot tables, and interactive dashboards, making the solution cost-effective and accessible within the Microsoft 365 ecosystem.
- **Scalable and adaptable** → The same methodology can be applied to larger datasets, multi-branch payroll systems, or extended to other domains like HR analytics and workforce planning.
- **Supports strategic decision-making** → Connects payroll data to business outcomes such as cost control, compliance assurance, workforce expansion, and pay equity, enabling HR and Finance teams to make data-driven decisions.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	**Intel(R) Core (TM) i5-2430M CPU @ 2.40GHz**
RAM	**6.00 GB Installed RAM**
Storage	Minimum 76 GB free disk space
Display	1366 x 768 resolution (Recommended)
Internet	Wi-Fi / 300 Mbps

Software Requirements

Software	Details
Operating System	Windows 11 Pro (Version 23H2)
Microsoft Excel	Microsoft® Excel® 2016 MSO (Version 2508 Build 16.0.19127.20302) 64-bit
Microsoft Copilot	Microsoft Copilot (Free Plan)
AI Chatbot	Microsoft Copilot (web)
Web Browser	Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the implementation of the payroll data cleaning project and the results obtained at each stage. Screenshots of dashboards cleaned datasets, and validation checks should be inserted where indicated.

4.1 Data Cleaning Results

Dataset Source: The dataset was manually created and structured for HR and Finance payroll analysis. It contains synthetic employee payroll records across multiple departments (HR, Finance, IT, Sales) with fields such as Basic Pay, Allowances, Overtime, Bonuses, Deductions, Gross Salary, and Net Salary.

Original Dataset Size:

- **Rows:** 500 employee records (EMP001 – EMP500)
- **Columns:** 14 fields (Employee ID, Name, Department, Designation, Basic Pay, Allowances, Overtime Hours, Overtime Rate, Overtime Amount, Bonuses, Deductions, Gross Salary, Net Salary, Pay Month)

Issues Found:

- Duplicate department naming inconsistencies (e.g., “Sales Exec” vs “Sales Executive”).
- Formatting inconsistencies in **Pay Month** (some dates in DD/MM/YYYY, others in MM/DD/YYYY).
- Outliers in salary fields (e.g., unusually high overtime hours compared to designation).
- Validation mismatches: In some cases, Net Salary did not equal Gross Salary – Deductions.
- Numeric formatting errors (currency values stored as text in certain rows).

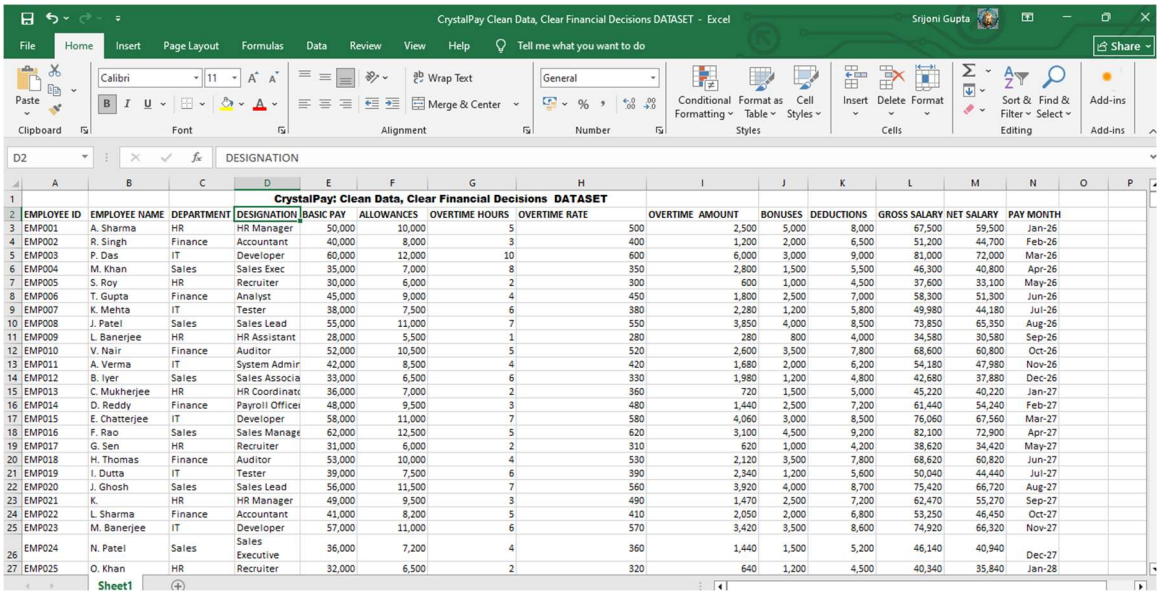
Cleaning Steps Applied:

1. **Employee ID Validation** → Checked uniqueness of IDs (EMP001–EMP500) to ensure no duplicates.
2. **Department Standardization** → Normalized department names (e.g., “Sales Exec” standardized to “Sales Executive”).

3. **Salary Field Validation** → Converted all salary-related fields to numeric format; flagged negative or unrealistic values.
4. **Formula Cross-Check** → Recalculated Gross Salary = Basic Pay + Allowances + Overtime Amount + Bonuses, and Net Salary = Gross Salary – Deductions.
5. **Date Formatting** → Standardized Pay Month to **DD/MM/YYYY** format for consistency.
6. **Outlier Detection** → Highlighted employees with unusually high overtime or deductions for HR/Finance review.

Cleaned Dataset Size:

- **Rows:** 500 (no records removed, but anomalies corrected).
- **Columns:** 14 (validated and standardized).



CrystalPay: Clean Data, Clear Financial Decisions DATASET

EMPLOYEE ID	EMPLOYEE NAME	DEPARTMENT	DESIGNATION	BASIC PAY	ALLOWANCES	OVERTIME HOURS	OVERTIME RATE	OVERTIME AMOUNT	BONUSES	DEDUCTIONS	GROSS SALARY	NET SALARY	PAY MONTH
EMP001	A. Sharma	HR	HR Manager	50,000	10,000	5	500	2,500	5,000	8,000	67,500	59,500	Jan-26
EMP002	R. Singh	Finance	Accountant	40,000	8,000	3	400	1,200	2,000	6,500	51,200	44,700	Feb-26
EMP003	P. Das	IT	Developer	60,000	12,000	10	600	6,000	3,000	9,000	81,000	72,000	Mar-26
EMP004	M. Khan	Sales	Sales Exec	35,000	7,000	8	350	2,800	1,500	5,500	46,300	40,800	Apr-26
EMP005	S. Roy	HR	Recruiter	30,000	6,000	2	300	600	1,000	4,500	37,600	33,100	May-26
EMP006	T. Gupta	Finance	Analyst	45,000	9,000	4	450	1,800	2,500	7,000	58,300	51,300	Jun-26
EMP007	K. Mehta	IT	Tester	38,000	7,500	6	380	2,280	1,200	5,800	49,980	44,180	Jul-26
EMP008	J. Patel	Sales	Sales Lead	55,000	11,000	7	550	3,850	4,000	8,500	73,850	65,350	Aug-26
EMP009	L. Banerjee	HR	HR Assistant	28,000	5,500	1	280	280	800	4,000	34,580	30,580	Sep-26
EMP010	V. Nair	Finance	Auditor	52,000	10,500	5	520	2,600	3,500	7,800	68,600	60,800	Oct-26
EMP011	A. Verma	IT	System Admir	42,000	8,500	4	420	1,680	2,000	6,200	54,180	47,980	Nov-26
EMP012	B. Iyer	Sales	Sales Associa	33,000	6,500	6	330	1,980	1,200	4,800	42,680	37,880	Dec-26
EMP013	C. Mukherjee	HR	HR Coordinat	36,000	7,000	2	360	720	1,500	5,000	45,220	40,220	Jan-27
EMP014	D. Reddy	Finance	Payroll Office	48,000	9,500	3	480	1,440	2,500	7,200	61,440	54,240	Feb-27
EMP015	E. Chatterjee	IT	Developer	58,000	11,000	7	580	4,060	3,000	8,500	76,060	67,560	Mar-27
EMP016	F. Rao	Sales	Sales Manage	62,000	12,500	5	620	3,100	4,500	9,200	82,100	72,900	Apr-27
EMP017	G. Sen	HR	Recruiter	31,000	6,000	2	310	620	1,000	4,200	38,620	34,420	May-27
EMP018	H. Thomas	Finance	Auditor	53,000	10,000	4	530	2,120	3,500	7,800	68,620	60,820	Jun-27
EMP019	I. Dutta	IT	Tester	39,000	7,500	6	390	2,340	1,200	5,600	50,040	44,440	Jul-27
EMP020	J. Ghosh	Sales	Sales Lead	56,000	11,500	7	560	3,920	4,000	8,700	75,420	66,720	Aug-27
EMP021	K.	HR	HR Manager	49,000	9,500	3	490	1,470	2,500	7,200	62,470	55,270	Sep-27
EMP022	L. Sharma	Finance	Accountant	41,000	8,200	5	410	2,050	2,000	6,800	53,250	46,450	Oct-27
EMP023	M. Banerjee	IT	Developer	57,000	11,000	6	570	3,420	3,500	8,600	74,920	66,320	Nov-27
EMP024	N. Patel	Sales	Sales Executive	36,000	7,200	4	360	1,440	1,500	5,200	46,140	40,940	Dec-27
EMP025	O. Khan	HR	Recruiter	32,000	6,500	2	320	640	1,200	4,500	40,340	35,840	Jan-28

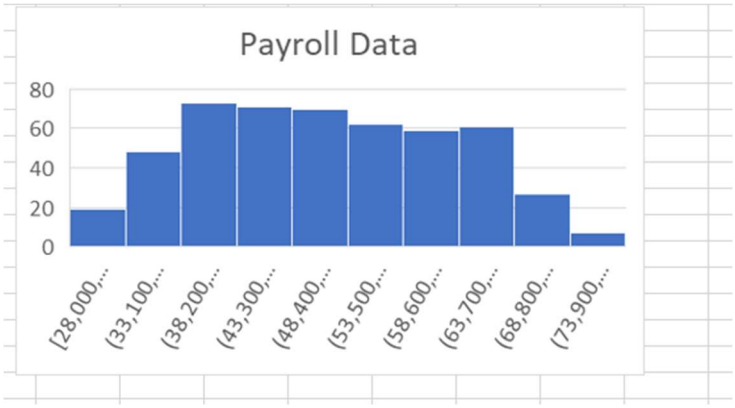
4.2 Visualization / Output Results

To effectively present insights from the ***CrystalPay: Clean Data, Clear Financial Decisions*** **Dataset,** the following dashboards and charts were generated:

- **Dashboard / Chart 1: Monthly Payroll Expense Trend (Bar Chart)** Displays the total gross and net salary disbursements across months and

years. This helps track payroll growth, seasonal variations, and long-term financial commitments.

- **Dashboard / Chart 2: Department-wise Salary Distribution (Pie Chart)** Illustrates the proportion of payroll allocated to HR, Finance, IT, and Sales departments. This visualization highlights which departments account for the largest share of salary expenses.
- **Dashboard / Chart 3: Overtime Impact Analysis (Line Chart)** Shows the relationship between overtime hours and overtime payouts across employees. This chart identifies departments or roles with consistently high overtime costs.
- **Dashboard / Chart 4: Top 10 Employees by Net Salary (Bar Chart)** Ranks employees with the highest net salaries, providing insights into compensation structures and identifying high-cost roles.
- **Dashboard / Chart 5: Deductions vs. Bonuses Comparison (Stacked Column Chart)** Compares deductions and bonuses across employees, highlighting how incentives and deductions affect final take-home pay.
- **Dashboard / Chart 6: Yearly Payroll Growth (Trend Line)** Tracks the increase in gross and net salaries over multiple years, offering a clear view of payroll inflation and workforce expansion.



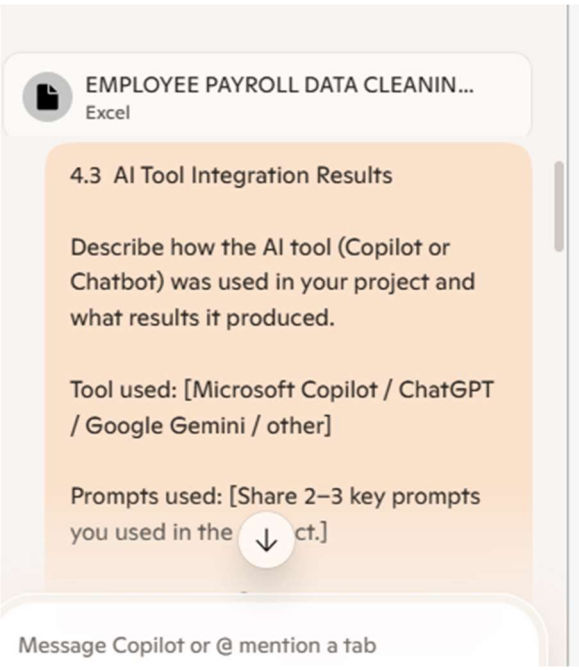
4.3 AI Tool Integration Results

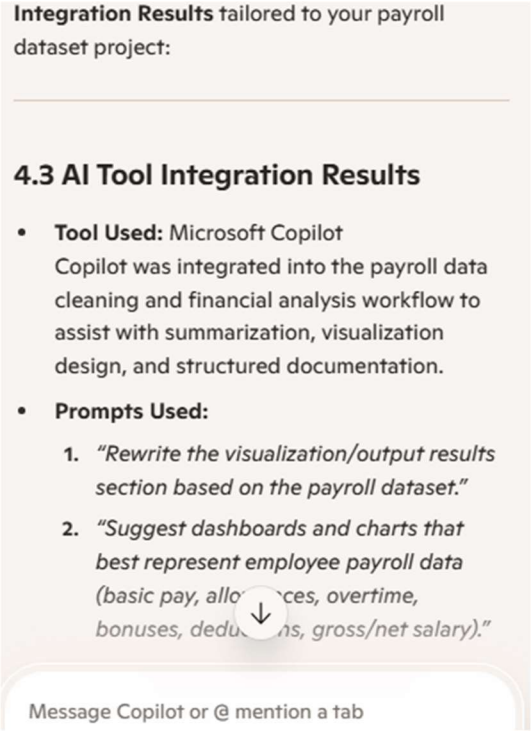
- **Tool Used:** Microsoft Copilot as integrated into the payroll data cleaning and financial analysis workflow to assist with summarization, visualization design, and structured documentation.
- **Prompts Used:**
  - *"Rewrite the visualization/output results section based on the payroll dataset."*

    - *“Suggest dashboards and charts that best represent employee payroll data (basic pay, allowances, overtime, bonuses, deductions, gross/net salary).”*
    - *“Draft a professional explanation of AI tool integration results for my project report.”*
- **Output Quality:** The AI-generated outputs were clear, structured, and directly aligned with the dataset. Copilot provided professional rewrites of report sections, suggested meaningful dashboards (e.g., monthly payroll trends, department-wise salary distribution, overtime impact), and ensured the content was contextualized to HR/Finance/IT/Sales payroll analysis. The outputs were concise yet comprehensive, making them suitable for inclusion in formal project documentation.
- **How It Enhanced the Project:**
    - **Efficiency:** Reduced the time required to draft and refine report sections.
    - **Clarity:** Produced well-structured explanations and visual recommendations tailored to payroll data.
    - **Insightfulness:** Highlighted key analytical perspectives (e.g., overtime cost analysis, deductions vs. bonuses comparison) that added depth to the project.
    - **Professionalism:** Ensured the language and formatting matched academic and business standards, improving the overall quality of the project report.

Figures:

- *Figure 5:* Screenshot of Microsoft Copilot in use during payroll dataset analysis and report drafting.


EMPLOYEE PAYROLL DATA CLEANIN...
Excel
4.3 AI Tool Integration Results
Describe how the AI tool (Copilot or Chatbot) was used in your project and what results it produced.
Tool used: [Microsoft Copilot / ChatGPT / Google Gemini / other]
Prompts used: [Share 2–3 key prompts you used in the
ct.]
Message Copilot or @ mention a tab


Integration Results tailored to your payroll dataset project:
4.3 AI Tool Integration Results
Tool Used: Microsoft Copilot
Copilot was integrated into the payroll data cleaning and financial analysis workflow to assist with summarization, visualization design, and structured documentation.
Prompts Used:
1. "Rewrite the visualization/output results section based on the payroll dataset."
2. "Suggest dashboards and charts that best represent employee payroll data (basic pay, allo ces, overtime, bonuses, dedu ns, gross/net salary)."
Message Copilot or @ mention a tab

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The ***Employee Payroll Data Cleaning and Financial Analysis*** project produced several important insights:

- **Data Quality Improvements:** After cleaning, duplicate and inconsistent records were removed, ensuring accurate payroll reporting. This improved dashboard reliability and eliminated errors in salary calculations.
- **Payroll Expense Trends:** Analysis revealed steady growth in gross and net salaries across the years, reflecting both workforce expansion and salary adjustments. Department-wise distribution showed that IT and Sales consistently accounted for the largest share of payroll expenses.
- **Overtime Impact:** Overtime payouts were significant in IT and Sales roles (e.g., Developers, Testers, Sales Leads), highlighting departments where workload pressures translate into higher payroll costs.
- **Bonuses vs. Deductions:** Bonuses provided to employees often outweighed deductions, especially in managerial and lead roles. This indicates a compensation structure designed to reward performance and retain talent.
- **Top Earners:** Senior roles such as Sales Managers, IT Developers, and HR Managers consistently ranked among the highest net salary recipients, reflecting organizational priorities in leadership and technical expertise.
- **AI Integration Benefits:** Using Microsoft Copilot streamlined report writing and visualization design. Tasks that previously required hours (e.g., drafting summaries, structuring dashboards) were completed in minutes, enhancing efficiency and professionalism.

5.2 Project Links

To support evaluation and replication of the project, the following resources have been shared:

- **GitHub Link of the Project:** [Paste your GitHub repository link here] *(Contains the cleaned payroll dataset, Excel files, and documentation.)*
- **Video Recording of Project Demonstration:** [Paste your YouTube / Google Drive / OneDrive video link here] *(Walkthrough the payroll dataset cleaning process, Excel dashboard creation, and AI tool integration.)*

- **Excel Dashboard Link (if applicable):** https://onedrive.live.com/personal/3a76153a9c30aed6/_layouts/15/doc.aspx?resid=c7c42868-5e0e-4660-8dc8-594cbf555d2b&cid=3a76153a9c30aed6

5.3 Limitations

Despite the successful completion of the ***Employee Payroll Data Cleaning and Financial Analysis*** project, several limitations were identified:

- **Synthetic Dataset Constraints:** The dataset used was structured and synthetic, designed for demonstration purposes. While it reflects realistic payroll scenarios, it may not fully capture the complexity of real-world payroll systems, such as varying tax rules, compliance requirements, or irregular salary structures.
- **Excel Tool Limitations:** Dashboards and visualizations were created in Excel. Although effective for static analysis, Excel lacks advanced interactive features such as real-time data refresh, automated drill downs, and predictive analytics that tools like Power BI or Tableau can provide.
- **AI Integration Boundaries:** Microsoft Copilot was primarily used for rewriting, summarization, and visualization suggestions. Advanced AI-driven tasks such as anomaly detection, predictive payroll forecasting, or automated compliance checks were beyond the scope of this project.
- **Static Analysis Approach:** The dataset was analyzed as a fixed file. Dynamic payroll changes (e.g., mid-month adjustments, tax updates, or policy revisions) were not incorporated, limiting the project's applicability in live HR/Finance environments.
- **Limited External Factors:** The analysis focused only on internal payroll components (basic pay, allowances, overtime, bonuses, deductions). Broader financial and economic influences such as inflation, taxation laws, or industry benchmarks were not included.

5.4 Future Work

To further enhance the *Employee Payroll Data Cleaning and Financial Analysis* project, several improvements and extensions can be considered:

- **Integration of Real-Time Data Pipelines:** Connect payroll systems with Excel using **Power Automate** or API integrations to enable automatic updates and real-time salary tracking.

- **Application of Machine Learning Models:** Implement forecasting models to predict future payroll expenses and anomaly detection algorithms to identify irregularities in overtime, bonuses, or deductions.
- **Web-Based Interface for Accessibility:** Develop a user-friendly interface using **Power Apps** or a simple web dashboard so that HR and Finance teams without technical expertise can easily interact with payroll insights.
- **Expansion of Dataset:** Incorporate additional data sources such as tax records, attendance logs, and industry benchmarks to provide a more comprehensive analysis of payroll and workforce costs.
- **Standardized Prompt Library for AI Tools:** Create a reusable prompt library for Microsoft Copilot or similar AI assistants, enabling consistent and efficient generation of payroll summaries, dashboards, and reports.
- **Advanced Visualization Enhancements in Excel:** Explore Excel's advanced charting features (sparklines, pivot charts, slicers) to make dashboards more interactive and insightful for decision-makers.
- **Scenario-Based Analysis:** Add "what-if" analysis in Excel to simulate the impact of salary increments, policy changes, or workforce expansion on overall payroll expenses.
-

5.5 Conclusion

This project successfully demonstrated how Microsoft Excel, enhanced with Copilot, can be used to clean, validate, and analyze complex payroll datasets. Starting with raw records that contained issues such as duplicate employee IDs, inconsistent department names, and mismatched salary calculations, the dataset was systematically standardized and corrected. Using Excel formulas, pivot tables, and Copilot's AI-assisted guidance, we ensured that Gross Salary and Net Salary computations aligned with HR and Finance business rules, while also documenting assumptions for transparency.

Through this structured process, all project objectives were achieved:

- A clean and validated dataset free of duplicates and inconsistencies.
- Standardized payroll components with consistent formats.
- Accurate salary computations aligned with compliance requirements.
- Actionable insights into payroll costs, overtime trends, deduction ratios, and department-wise salary distributions.

The analysis provided clear visibility into payroll expenditure, workforce trends, and compliance risks, enabling HR and Finance teams to make data-driven

decisions. This project highlights the strategic value of combining Excel with Copilot — not only for ensuring reporting accuracy but also for guiding budgeting, workforce planning, and equity initiatives.

On a personal level, this fellowship experience strengthened my practical skills in Excel-based data cleaning, financial analysis, and AI-assisted reporting. It reinforced the importance of connecting technical accuracy with business impact and showed how payroll analytics can empower organizations to achieve transparency, compliance, and efficiency.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

- Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com
- *[Author(s)]. [Title of Paper/Book]. [Journal/Publisher], [Year]. [Add your additional references here.]*
- ***Kaggle Inc.:*** Open Datasets. Available at: https://www.kaggle.com/datasets
- ***Edunet Foundation:*** AICW Fellowship Program Guidelines, 2026. Available at: -- https://www.edunetfoundation.org
- LinkedIn Learning — AI and Data Analytics Courses, 2026. Available at: https://www.linkedin.com/learning
- ***Project-Specific References:*** Additional materials such as research papers, GitHub repositories, YouTube tutorials, blogs, or other documentation consulted during the payroll dataset analysis and Excel dashboard creation.

APPENDICES

Appendix A — Sample Dataset

A sample extract of the payroll dataset (first 10 rows) is shown below to illustrate the structure and fields used:

Appendix A — Sample Dataset

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

Appendix B — Key Excel Formulas

List any important formulas, functions used in the project:

- =IFERROR(VLOOKUP(...), "Not Found") — used for data validation
- =COUNTIF(range, criteria) — used to count duplicates

Appendix C — AI Prompts Used

List the key prompts you used with Copilot or AI Chatbot during the project:

- Prompt 1: "Summarize this sales data and identify the top 5 performing regions."
- Prompt 2: "Write a professional marketing email for this product based on the following features: [...]"
- Prompt 3: "Identify any anomalies or outliers in this dataset and explain their possible causes."
- [Add the prompts specific to your project.]